

Richa Patel

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience in Document AI, Machine Learning, and NLP, specializing in large datasets, predictive modeling, data visualization, and data acquisition. Expertise in building end-to-end data pipelines, model development, and cloud technologies (GCP, AWS). Proficient in Python, TensorFlow, LangChain, and OpenAI APIs, with a strong focus on optimizing AI-driven document processing and delivering scalable solutions for data-driven business outcomes. Passionate about leveraging generative AI and NLP to enhance operational efficiency and model performance.

WORK EXPERIENCE

Data Engineer | Cleardox LLC (Stamford, CT)

Sep 2022 – Jan 2025

- **Designed and deployed scalable Document AI Pipelines** using Vertex AI, TensorFlow, and BigQuery for OCR, data extraction, and document classification, enabling efficient large-scale document processing.
- **Developed advanced semantic chunking and entity extraction algorithms**, enhancing Named Entity Recognition (NER) and improving document classification accuracy.
- **Integrated Pytest for end-to-end testing automation**, covering document annotations, chunking, embeddings, and metadata extraction, increasing testing efficiency by 40%.
- **Led the development of AI-powered search and retrieval systems** using PG Vector, LangChain, and Gemini LLM models, ensuring high-speed and reliable document retrieval.
- **Optimized RAG (Retrieval-Augmented Generation) pipelines**, enhancing data retrieval, embedding generation, and improving performance for document comparison and search.
- **Built and deployed API solutions for document classification and retrieval**, ensuring automated validation through rigorous testing for accuracy and consistency.
- **Implemented dynamic prompt engineering techniques** to optimize LLM model performance, improving response accuracy and relevance for search and classification tasks.

Data Engineer | NewTech Engineers (Gujarat, India)

May 2018 – Nov 2020

- **Developed and optimized scalable ETL pipelines** using AWS services (S3, Lambda, Glue, Athena, Databricks), facilitating large-scale data processing.
- **Implemented event-based real-time data integration pipelines in Golang**, significantly improving third-party integrations and reducing data processing delays.
- **Integrated DynamoDB with AWS services** (Lambda, API Gateway, S3) to ensure seamless data flow and improved system reliability.
- **Built Python microservices with Django** to streamline data handling and integration with front-end applications.
- **Developed interactive Business Intelligence dashboards** using Google Looker, enabling real-time trend analysis and data-driven decision-making.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, R
- **Cloud Technologies:** Google Cloud Platform (GCP), AWS (Lambda, Glue, S3, EC2, BigQuery)
- **AI/ML:** Natural Language Processing (NLP), Named Entity Recognition (NER), Semantic Chunking, TensorFlow, LangChain, LlamaIndex, Generative AI (LLM models), Retrieval-Augmented Generation (RAG), Text Classification
- **Data Engineering:** Kafka, PostgreSQL, MySQL, Hive, Databricks, Data Lakes, ETL Pipelines
- **Business Intelligence:** Looker, Power BI, Tableau
- **Testing & Automation:** Pytest (unit tests, regression tests, mocking, CI/CD integration), API Testing
- **Packages:** Pandas, NumPy, ScikitLearn, PyTorch, TensorFlow, Keras, Matplotlib
- **Tools & Operating Systems:** Git, MS Excel, Windows, Linux

EDUCATION

Master of Science in Data Science

University of New Haven | Jan 2021 – Aug 2022

Master of Science in Applied Mathematics

Maharaja Sayajirao University, Gujarat, India | June 2017 - May 2019

CERTIFICATIONS

- Pytest Testcase – Pluralsight
- Python for Data Science and Machine Learning Bootcamp – Udemy

PROJECTS

1. NLP Annotation and Document AI Pipeline Implementation

- Designed and implemented a **scalable Document AI pipeline** for OCR, data extraction, and classification, leveraging Vertex AI, TensorFlow, and BigQuery.
- Developed **semantic chunking and text entity extraction algorithms**, improving Named Entity Recognition (NER) and document classification accuracy.
- Integrated **Pytest for automated testing**, covering document annotations, chunking, embeddings, and metadata extraction, boosting testing efficiency by 40%.
- Optimized data ingestion, storage, and model deployment workflows using **Google Cloud Platform (GCP)**, enhancing system performance by 30%.

2. Generative AI and Vector Database for Document Search and Retrieval

- Led the development of an **AI-powered document search and retrieval application** using PG Vector, LangChain, and Gemini 1.5 Pro (LLM).
- Designed **Retrieval-Augmented Generation (RAG) pipelines** with LlamaIndex and PG Vector, optimizing data retrieval and embedding generation.
- Implemented **dynamic prompt engineering techniques** to fine-tune LLM performance for document classification, search, and content generation.
- Conducted **API development and integration testing**, including endpoint validation, LLM chunking, and document comparison for robust system performance.

3. MLOps Pipeline for Model Training, Evaluation, and Deployment

- Developed an **MLOps pipeline using GCP tools (Vertex AI, BigQuery)** for automating model training, evaluation, and deployment.
- Implemented **fine-tuning and model versioning strategies** for NLP-based models (NER, text classification), leveraging **fuzzy matching techniques** for validation.
- Built a **comprehensive Pytest suite** to validate model outputs, covering edge cases and regression tests, ensuring high-quality performance.